

FNU TRUPTI

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SUMMARY

Data Analyst with over 2 years of experience designing and automating data pipelines, building models, and delivering actionable insights. Skilled in Python, SQL, Pandas, Power BI, and statistical modeling; developed ETL pipelines that cut reporting time by 40% and crafted dashboards for real-time monitoring. Proven ability to interpret complex datasets to drive decision-making.

EDUCATION

University of Maryland, Baltimore County

May 2025

Master's, Information Systems

- GPA: 3.9

Godutai Engineering College for Women

Aug 2021

Bachelor's, Electronics and Communication

- GPA: 8.9

EXPERIENCE

NEOPATHOLOGY CORP.

Research And Development Intern

Jun 2024 - Aug 2024

- Developed Python feature extraction algorithms using `skimage.measure.regionprops` to quantify shapes in tissue structures.
- Conducted literature reviews and refined morphological features using statistical modeling to ensure robust and biologically meaningful shape analysis.
- Collaborated cross-functionally to validate shape descriptors and improve measurement completeness by 26%.
- Maintained reproducible analysis pipelines using Git, ensuring consistent results across model iterations.

UNIVERSITY OF MARYLAND, BALTIMORE COUNTY

Research Assistant

Feb 2024 - Sep 2025

- Designed statistical and predictive data models using survival analysis and fairness algorithms for equitable healthcare risk prediction.
- Engineered fairness metrics in Python (e.g., demographic parity, equal opportunity) to reduce bias in healthcare resource allocation.
- Built data pipelines with Pandas and NumPy to clean and structure healthcare datasets (100K+ records).

Research Assistant

Sep 2024 - Nov 2025

- Developed multimodal dementia assessment model combining BERT (text) and Whisper (speech) for cognitive score prediction.
- Designed fusion network to integrate linguistic features and acoustic biomarkers from clinical speech data.
- Implemented multi-GPU training with hyperparameter optimization for enhanced model performance.
- Evaluated using RMSE, MAE, and correlation metrics, achieving strong predictive accuracy.

GOVERNMENT TOOL ROOM AND TRAINING CENTER

Data Analyst

Mar 2021 - May 2023

- Developed SQL systems, data models, and automated Python ETL pipelines to monitor machine KPIs, reducing manual reporting time by 40%.
- Designed Power BI dashboards and generated detailed KPI reports to monitor utilization, downtime, and throughput, improving visibility and accelerating decisions.
- Leveraged consulting experience to integrate PLC, SCADA, and IIoT via MQTT and Raspberry Pi for predictive maintenance, cutting failures by 15%.
- Conducted time-series forecasting with CMMS data in Azure Data Studio, increasing production throughput by 12% and reducing machine downtime by 10%.
- Integrated data from SCADA and CMMS into centralized dashboards for real-time and historical insights.

SKILLS

- **Languages:** Python, SQL, R, Java, C/C++, HTML, CSS, JavaScript, VHDL, VLSI
- **Data Analysis & ML:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Predictive Analytics, Statistical Modeling, Time Series Forecasting, Ad-hoc Analysis, A/B Testing, Data Annotation, Quality Assurance, Quality Auditing, Data Modeling
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn, MS Excel (Advanced Functions, Pivot Table, VBA, Macros), Streamlit, KPI Dashboards, Power Query, GIS
- **Databases:** MySQL, Oracle, SQL Server, MongoDB, NoSQL, Snowflake
- **Big Data & Cloud:** Databricks, AWS, Azure, GCP, CI/CD, Hadoop, Docker, GitHub, JIRA
- **Specialized Tools:** PLC, SCADA, CMMS, MQTT, Raspberry Pi, Adobe XD, Lens Studio
- **Industry Expertise:** Tech Industry Experience, Consulting Experience

PROJECT

Customer Accounts Operations Project

Jan 2026

- Created a consolidated Accounts Receivable (AR) master dataset by integrating customer, invoice, payment, and interaction data to support Power BI dashboards and financial reporting.
- Automated key business metrics including aging buckets, Days Sales Outstanding (DSO), outstanding balances, and payment cycle analysis to improve collections efficiency and customer risk assessment.

Spotify AI Recommender

Apr 2025

- Built a restaurant management system in PL/SQL with normalized schemas, stored procedures, triggers, and automated reporting.
- Integrated EDA and PCA visualizations for playlist analysis, collected user feedback for UX improvement.

Database Management System

Oct 2023

- Built a restaurant management system in PL/SQL with normalized schemas, stored procedures, triggers, and automated reporting.
- Modeled and optimized schemas using MySQL Workbench, enabling real-time insights into sales, inventory, and order trends.